Limits

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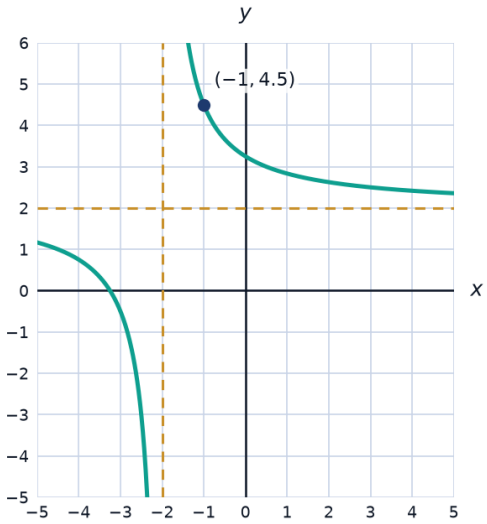
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Explain why you cannot find $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$ by direct substitution alone.

2 **[Foundational]** • **Target: Limits at infinity for rational functions**

Explain what happens to $\lim_{x \to \infty} \frac{5x^3 - 2}{x^2 + 1}$, without computing an exact value.



3 **[Foundational]** • **Target: Limits at infinity for rational functions**

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

4 **[Foundational]** • **Target: Direct substitution**

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

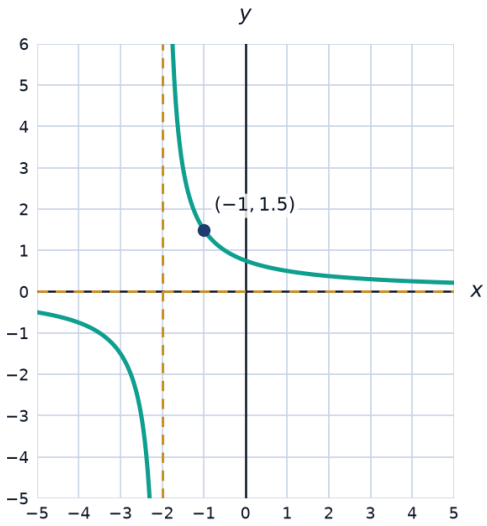
5 **[Foundational]** • **Target: Indeterminate forms: factor and cancel**

Solve the system using matrix inversion: $4x + 7y = 11$ $x + 2y = 3$

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6 [Foundational] • Target: Limits at infinity for rational functions

Find all real solutions to the radical equation: $\sqrt{2x + 15} - x = 0$



7 [Foundational] • Target: Direct substitution

In a geometric sequence, $a_1 = 5$ and $a_4 = 40$. Find the common ratio r and the 7th term a_7.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Indeterminate forms: factor and cancel

Calculate the determinant of the matrix $M = (7, 3; -2, 5)$.

9 [Intermediate] • Target: Limits at infinity for rational functions

Convert the repeating decimal $0.\overline{45}$ to an irreducible common fraction.

10 [Intermediate] • Target: Direct substitution

Find the 25th term of the arithmetic sequence: $7, 12, 17, 22, \ldots$

11 [Intermediate] • Target: Indeterminate forms: factor and cancel

In triangle ABC, sides are $a = 5$, $b = 6$, and $c = 7$. Find the exact value of $\cos A$.

12 [Intermediate] • Target: Limits at infinity for rational functions

If $z_1 = 5 - 2i$ and $z_2 = -3 + 7i$, find the value of $2z_1 - z_2$ in standard form $a + bi$.

13 [Intermediate] • Target: Direct substitution

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

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14 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find all real solutions to the rational equation $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Decompose the rational expression $R(x) = \frac{7x-1}{x^2-x-6}$ into partial fractions using the Heaviside cover-up method.

16 [Challenging] • Target: Direct substitution • Cross-Lesson: 1.01 Complex Numbers

Let $f(x) = \frac{x^2-9}{x^2-5x+6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

17 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 2.01 Matrices & Vectors

Convert the repeating decimal $x = 0.4\overline{36} = 0.4363636\ldots$ into an irreducible rational fraction $\frac{p}{q}$.

18 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 3.01 Sequences & Series

Determine whether the function $g(x) = \frac{x^3}{x^2+4}$ is even, odd, or neither.

19 [Challenging] • Target: Direct substitution • Cross-Lesson: 4.01 Advanced Functions & Equations

If $f(x) = 2x - 5$ and $g(x) = x^2 + 3$, find the expression for $(f \circ g)(x)$ and solve for all x such that $(f \circ g)(x) = 25$.

20 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Direct substitution

Find $\lim_{x\to\infty} \frac{x^3+5x}{9x^2-9x}$ (state whether it is ∞ or $-\infty$).

a. $\frac{1}{9}$

b. 0

c. $-\infty$

d. ∞

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22 [Foundational] • Target: Indeterminate forms: factor and cancel

For real numbers a and b, suppose that $\lim_{x\to 1} \frac{x^2+ax+b}{x^2-1} = 3$. What is the value of $a-b$?

a. -9

b. -1

c. 3

d. 9

23 [Foundational] • Target: Limits at infinity for rational functions

What is the value of the limit $\lim_{x\to 2} \frac{x^2+x-6}{x-2}$?

a. 5

b. 1

c. -1

d. Does not exist

24 [Foundational] • Target: Direct substitution

For $f(x) = 2x^3 + 2x^2 - x + 6$, which values are possible rational roots (by the Rational Root Theorem)?

a. $\pm 1, \pm 2$

b. ± 6

c. ± 12

d. $\pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \pm 3, \pm 6$

25 [Foundational] • Target: Indeterminate forms: factor and cancel

Find all vertical asymptotes of $f(x) = \frac{x^2+x-6}{x^3-6x^2+3x+10}$.

a. x=2,-1,5

b. x=2

c. x=-1

d. x=-1,5

26 [Foundational] • Target: Limits at infinity for rational functions

For $f(x) = \frac{(x+2)(x+1)}{(x+1)(x+3)}$, identify the hole and vertical asymptote.

a. Hole at $x = -3$, vertical asymptote at $x = -1$

b. Hole at $x = -1$, vertical asymptote at $x = -3$

c. Vertical asymptotes at $x = -1$ and $x = -3$, no hole

d. Hole at $x = -2$, vertical asymptote at $x = -3$

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27 [Foundational] • Target: Direct substitution

For $f(x) = (x - 4)^3(x + 10)$, describe the behavior of the graph at $x = 4$.

a. crosses the x-axis normally (linear)

b. crosses the x-axis, flattening (S-shaped) near the zero

c. touches the x-axis and turns around (does not cross)

d. the graph has a vertical asymptote there

28 [Foundational] • Target: Indeterminate forms: factor and cancel

As $x \to +\infty$, what happens to $f(x) = 2x^5$?

a. $+\infty$

b. $-\infty$

c. 0

d. Oscillates

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Limits at infinity for rational functions

Find the slant (oblique) asymptote of $f(x) = \frac{x^2 + 5x + 1}{x - 1}$.

a. $y = 6x$

b. $y = x + 6$

c. $y = x + 7$

d. No slant asymptote

30 [Intermediate] • Target: Direct substitution

Find the x-coordinate of the hole in the graph of $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$.

a. x=5

b. x=2

c. No hole exists

d. x=-3

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31 [Intermediate] • Target: Indeterminate forms: factor and cancel

Find the horizontal asymptote of $f(x) = \frac{x^3+1}{x^2-4}$, or state that none exists.

a. The line $y = x$

b. y=0

c. y=1

d. No horizontal asymptote

32 [Intermediate] • Target: Limits at infinity for rational functions

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. 27

d. 26

33 [Intermediate] • Target: Direct substitution

What is the horizontal asymptote of $f(x) = \frac{1}{x}$?

a. $y = 1$

b. $y = 0$

c. None

d. $x = 0$

34 [Intermediate] • Target: Indeterminate forms: factor and cancel

What is the vertical asymptote of $f(x) = \frac{1}{x-0}$?

a. $x = 0$

b. $y = 0$

c. $x = 1$

d. None

35 [Intermediate] • Target: Limits at infinity for rational functions

Find the horizontal asymptote of $f(x) = \frac{3}{x^2-4}$.

a. y=0

b. y=-1

c. y=1

d. y=3

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36 [Intermediate] • Target: Direct substitution

Simplify $f(x) = \frac{x^2 + x - 6}{x^2 - 7x + 10}$ and evaluate at $x = 1$.

a. 1

b. 0

c. 2

d. -1

37 [Intermediate] • Target: Indeterminate forms: factor and cancel

What is the degree and leading coefficient of $f(x) = 2x^4 + x^2 - 2$?

a. Degree 5, leading coefficient 2

b. Degree 4, leading coefficient 2

c. Degree 4, leading coefficient -2

d. Degree 3, leading coefficient 2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 3.01 Sequences & Series

Find the vertical asymptote of $f(x) = \frac{2}{x - 3}$.

a. y=3

b. x=0

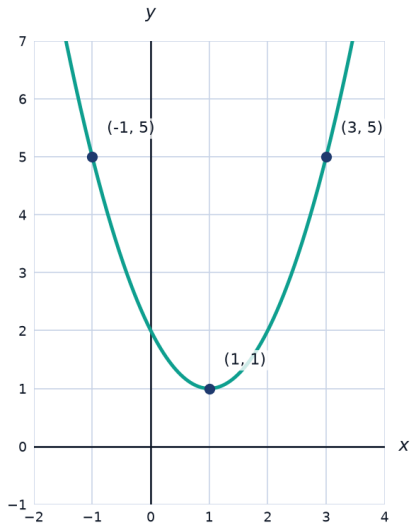
c. x=3

d. x=-3

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39 [Challenging] • Target: Direct substitution • Cross-Lesson: 4.01 Advanced Functions & Equations

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.



a. 0

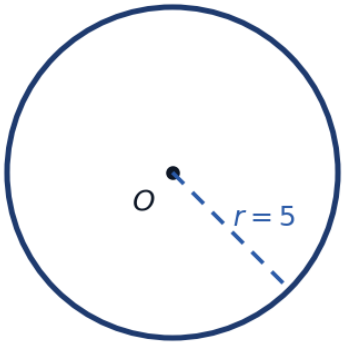
b. -2

c. 2

d. 1

40 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

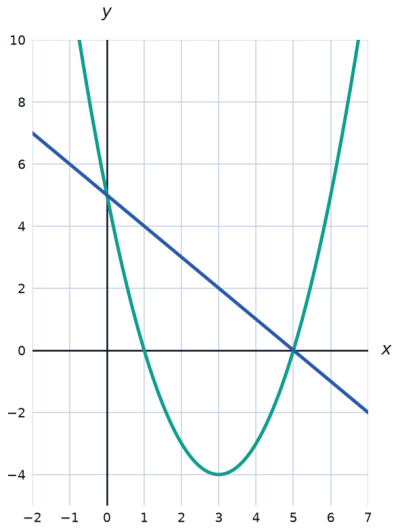
c. 1

d. $\frac{\sqrt{2}}{2}$

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41 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 1.01 Complex Numbers

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?



a. $\sqrt{16}$

b. $\sqrt{25}$

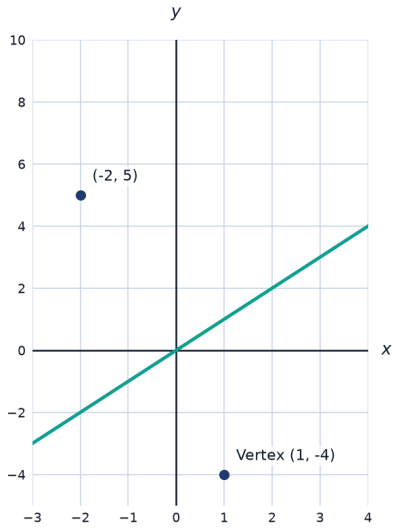
c. $\sqrt{36}$

d. $\sqrt{49}$

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42 [Challenging] • Target: Direct substitution • Cross-Lesson: 2.01 Matrices & Vectors

The graph of the quadratic function $f(x) = x^2 + bx + c$ is shown below. The graph passes through the point $(-2, 5)$ and has its vertex at $(1, -4)$. Determine the value of b.



a. 1

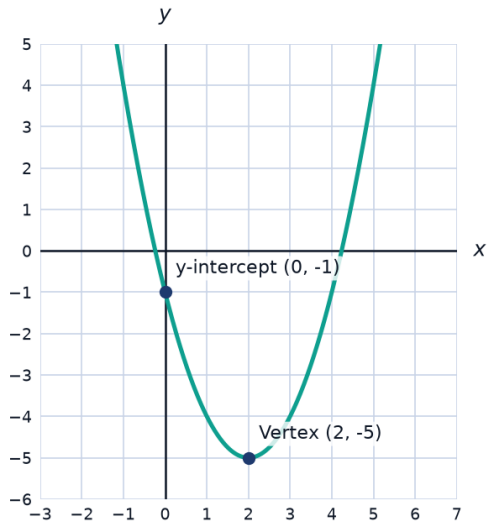
b. -2

c. -1

d. 2

43 [Challenging] • Target: Indeterminate forms: factor and cancel • Cross-Lesson: 3.01 Sequences & Series

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at $(2, -5)$ and the y-intercept is at $(0, -1)$. Determine the value of a.



a. 1

b. -1

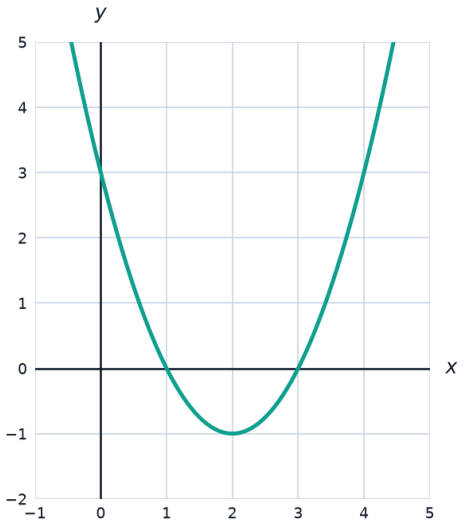
c. 2

d. -2

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44 [Challenging] • Target: Limits at infinity for rational functions • Cross-Lesson: 4.01 Advanced Functions & Equations

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?



a. (2, -1)

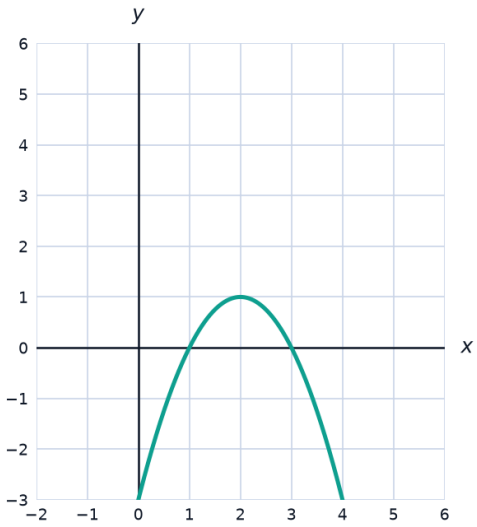
b. (2, 1)

c. (1, 0)

d. (3, 0)

45 [Challenging] • Target: Direct substitution • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the y-coordinate of the vertex of the parabola.



a. 1

b. 2

c. 3

d. 4

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